Exercise 1

SQL Fundamentals.

Select Statement

```
SELECT *
FROM employees ;
```

id	first_name	last_name	department	salary	hire date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-2004	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. Select DISTINCT statement : Write a SQL query to find all unique departments in the employees table.

```
SELECT DISTNCT department
FROM employees-db. schema. employees;
```

department
IT
HR
Finance
Marketing

3. ORDER BY statement : Write a SQL query to retrieve all employees' first and last names, ordered by salary in descending order

```
SELECT first-name, last-name, salary
FROM employees-db, schema, employees
ORDER BY salary ;
```

first_name	last-name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50 000
Jane	Smith	48 000

4 Limit statement : Write a query to retrieve the top 5 highest paid employees.

```
SELECT first_name
       last_name
       salary
FROM employees_db. schema. employees
ORDER BY salary DESC
LIMIT 5;
```

first_name	last_name	salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000

5. WHERE statement : Write a query to find employees who work in the IT department.

```
SELECT first_name,
       last_name
FROM employees_db. schema. employees
WHERE department = 'IT'
```

first_name	last_name
John	Doe
Sarah	Brown
Emily	Davis
Laura	Hall

6 Write a SQL query to find employees who work in the Finance department AND have a salary greater than 58000
AND statement

```
SELECT first_name
       last_name
FROM employees_db. schema. employees
WHERE department = 'IT' AND
      salary > 58000
```

first_name	last_name	salary
Emily	Davis	62000

7. OR Statements: Write a SQL query to find employees who work in the HR department OR the Marketing Department.

```
SELECT first_name,
       last_name
FROM employees_db.schema.employees
WHERE department = 'HR'
  OR  department = 'Marketing';
```

first_name	last_name
Jane	Smith
Jessica	Moore
David	White
Daniel	Clark

Write a SQL query to find employees who ~~are in HR, IT or Finance department~~ do not work in IT department.

8. NOT Statement

```
SELECT first_name,
       last_name, department
FROM employees_db.schema.employees
WHERE department NOT IN ('IT')
```

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

Write a SQL query to find employees who are in HR, IT, or Finance department

9. IN Statement

```
SELECT first_name,
       last_name
FROM employees_db.schema.employees
WHERE department IN (HR, IT, Finance)
```

first_name	last_name
John	Doe
Jane	Smith
Mike	Johnson
Sarah	Brown
Emily	Davis
Robert	Wilson
Jessica	Moore
Laura	Hill

Write a SQL query to find employees who are in the IT department have a salary greater than 50,000 and are located in New York

10 Combining conditions

```
SELECT first_name,
       last_name,
       salary,
       location
FROM employees_db.schema.employees
WHERE department = 'IT'
  AND salary > 50000
  AND city = 'New York'
```

first_name	last_name	salary	location
John	Doe	50000	New York

18 NOT Statement

Write a SQL query to find employees who do not work in the IT department

```
SELECT first_name,
       last_name
FROM
WHERE NOT department = 'IT'
```

first_name	last_name
Jane	Smith
Mike	Johnson
David	White
Robert	Wilson
Jessica	Moore
Daniel	Clark

Combining conditions

Write a query to find employees who are in the IT department, have a salary greater than 50,000 and are located in New York

```
SELECT first_name,
       last_name
FROM
WHERE department = 'IT',
      AND salary > 50000
      AND city = 'New York'
```

first_name	last_name
John	Doe
Sarah	Brown

Write a SQL query to retrieve the last and first of employees who work in the Finance or Marketing department earn more than $2,000, and order the result by salary in descending order.

```
SELECT first_name,
       lastname, department, salary
FROM employees_db, schema. employees
WHERE department = 'Finance' OR department = 'Marketing'
      AND salary > 52 000
ORDER BY salary DESC;
```

first_name	last_name	department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53 000

Write a SQL query to find all unique cities where employees work, excluding those in the IT and HR departments.

```
SELECT DISTINCT department
FROM employees_db. schema employees
WHERE department IN ( Marketing, Finance )
```

city
Los Angeles
San Francisco
Houston
Chicago

Write a SQL query to retrieve employees who are NOT in Finance department, have a salary greater than 50,000, and order the result by hire date in ascending order

```
SELECT*
FROM employees_db. schema. employees
WHERE NOT department = 'Finance'
        AND salary > 50000
ORDER BY hire_date;
```

id	first_name	last_name	department	salary	hire_date
6	Emily	Davis	IT	62000	2016-02-14
5	David	White	Marketing	57000	2016-04-10
8	Jessica	Moore	HR	51000	2018-05-22
1	Doe	John	IT	55000	2018-06-15
10	Laura	Hall	IT	53000	2020-08-10
4	Sara	Brown	IT	53000	2021-03-25
9	Daniel	Clark	Marketing	53000	2022-06-01